

Composite Flat Admissibility: The Gradient-Composite Scheme Admits the Affine Representer for All Coefficients

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Abstract

The effective Palatini variation of the minimal Hilbert–Palatini + kinetic branch excludes the flat affine representer (flat affine no-go, **GAP-10T-FLAT-NOGO**). We prove the complementary positive statement: in the *gradient-composite* torsion-free scheme, where the tetrad is the slice coordinate map of $\partial_\mu \Theta / \sqrt{\mathcal{N}_0}$ and only Θ is varied, the affine representer Θ_{aff} is a stationary point of the full action for *all* values of Λ , κ , and $\mathcal{N}_0$. The two variational schemes are therefore dynamically inequivalent at the flat point, and the Palatini no-go does not transfer. This closes **GAP-10T-COMPOSITE-FLAT** [L1] and sharpens **GAP-10T-DYN**: the surviving minimal continuation is composite.

1 Scheme and statement

Work on Lorentz-real fields: $\partial_\mu \Theta \in W_L$, tetrad e^a_μ defined by the slice coordinates of $\partial_\mu \Theta / \sqrt{\mathcal{N}_0}$, metric by the central Jordan identity, spin connection $\hat{\omega}(e)$ Levi–Civita (torsion-free branch), potential $V \equiv 0$. The action is

$$S[\Theta] = \frac{1}{4\kappa} \int \epsilon_{abcd} e^a \wedge e^b \wedge R^{cd}(\hat{\omega}(e)) - \frac{\Lambda}{2\kappa} \int \sqrt{-g} + \frac{1}{2} \int \sqrt{-g} g^{\mu\nu} \langle \partial_\mu \Theta, \partial_\nu \Theta \rangle_\# . \quad (1)$$

Only Θ varies; e , g , $\sqrt{-g}$, and $\hat{\omega}$ vary by the chain rule. This is the well-defined gradient-composite scheme; the self-consistent D -composite scheme (connection inside the defining jet) remains open and is *not* treated here.

Theorem 1 (Composite flat admissibility). $\Theta_{\text{aff}} = \Theta_0 + \sqrt{\mathcal{N}_0} \bar{E}_\mu x^\mu$ is a stationary point of (??) under compactly supported Lorentz-real variations, for all $(\Lambda, \kappa, \mathcal{N}_0)$.

Proof. The Lagrangian density of (??) is a smooth local function of the 2-jet of Θ in a neighbourhood of the affine jet (the affine tetrad is nondegenerate, so $\hat{\omega}(e)$ is smooth there), and it contains no undifferentiated Θ ($V \equiv 0$; the scheme replaces D_μ by ∂_μ in the defining jet). The Euler–Lagrange derivative is

$$\mathcal{E} = \frac{\partial L}{\partial \Theta} - \partial_\mu \frac{\partial L}{\partial (\partial_\mu \Theta)} + \partial_\mu \partial_\nu \frac{\partial L}{\partial (\partial_\mu \partial_\nu \Theta)}.$$

At the affine configuration the 2-jet is constant in x , so both momenta are constant matrices and their derivatives vanish; the first term vanishes because L has no undifferentiated Θ . Hence $\mathcal{E} = 0$.

Concretely, the three pieces vanish for transparent reasons: the kinetic and Λ terms are volume-type functionals of the induced metric, and their first variation at the constant background is $\propto \int \eta^{\mu\nu} \langle \bar{E}_\mu, \partial_\nu \delta \Theta \rangle_\# = - \int \langle \partial_\nu \bar{E}^\nu, \delta \Theta \rangle_\# = 0$; the Einstein term satisfies $R(\hat{\omega}(\bar{e})) = 0$ and $\delta \int \epsilon e e e R = \int \epsilon e e d(\delta \hat{\omega}) = \int d(\epsilon e e \delta \hat{\omega}) = 0$ since $de = 0$ for the constant background tetrad.

Both integral statements are verified exactly (SymPy, polynomial boundary-vanishing variations with linear and quadratic coefficients, linearised Levi–Civita connection) in `tools/verify_canonical_spin_cur` check C7. $\square$

Corollary 1 (Scheme inequivalence at the flat point). *The effective Palatini scheme excludes every flat affine representer (GAP-10T-FLAT-NOGO), while the gradient-composite scheme admits it for all coefficients. The two variational schemes of the same minimal action are dynamically inequivalent, and the surviving minimal continuation of canonical UBT is the composite one.*

Scope and non-claims

Proved for the gradient-composite torsion-free scheme with $V \equiv 0$ and Lorentz-real variations. Not proved: the self-consistent D -composite variation, stationarity on curved backgrounds, uniqueness or stability of the flat point, any derivation of S_{HP} or its coefficients. GAP-10T-DYN and GAP-10D remain NARROWED. Note the Λ -independence of flat stationarity: the composite scheme imposes only divergence-type conditions on covariantly constant metric-sector mismatches, a unimodular-like feature that must be treated explicitly in the curved dynamical analysis.